Piyush Bhamare

+1(617)-590-0193 | Boston, MA | piyushbhamare2111@gmail.com | LinkedIn

SUMMARY

Mechanical Engineer with hands-on experience in manufacturing process optimization, injection molding, quality control, and Lean Manufacturing. Skilled in SPC, root cause analysis, GD&T, and data analysis using MATLAB, Minitab, Excel, and SolidWorks. Proven ability to improve production efficiency, reduce rework, enhance product quality, and implement data-driven manufacturing solutions

EXPERIENCE

Manufacturing Engineer – Level 1| Mylan Laboratories India Pvt Ltd. | Nashik, India **Nov 2022 – Jul 2024**

- Calibrated Cognex/Keyence vision systems, improving SMT inspection accuracy by 15% and eliminating 300+ false rejects/month
- Diagnosed critical defects using 8D, Fishbone, 5 Whys, and video analysis, improving OEE 5% and preventing recurrence
- Integrated MES/SCADA with Power BI across 8 assembly lines for real-time defect tracking and root cause analysis
- Investigated 30+ NCMRs and implemented CAPAs using root cause analysis, reducing scrap and delivering $87K in annual savings
- Redesigned production fixtures using SolidWorks, DFM/DFA, and GD&T, increasing throughput by 80 units/day across 30+ SKUs
- Validated inspection systems using Gage R&R, TMV, Minitab, improving measurement reliability by 6% and ensuring FDA compliance
- Assisted process validation (IQ/OQ/PQ) across 4 production lines, enabling smooth transition from pilot to full-scale production

Manufacturing Engineering Intern | Shivkrupa Techno-Plast Ltd**.** | Nashik, India **Jan 2022 – Aug 2022**

- Monitored 11 injection-molding machines (450T–850T), using MATLAB to optimize process parameters and increase output by 20%
- Conducted time studies across finishing stations and redesigned trimming fixtures, reducing cycle time by 7 min/unit
- Monitored SPC control charts in Minitab, evaluating Cp/Cpk indices across 25+ molds to stabilize dimensional tolerances
- Standardized raw-material mixing and established SOPs using Lean and SPC tools, improving line balance and reducing rework by 25%
- Executed in-process quality inspections and Layered Process Audits (LPAs) to identify line fallout, reducing defect escape rates by 15%
- Analyzed production data to identify variation and improve process consistency and dimensional quality

Manufacturing Engineering Intern | Adamant Engineering PVT Ltd. | Nashik, India **Jan 2021 – Aug 2021**

- Improved FRP pultrusion stability with ANSYS FEA validation, adjusting resin curing parameters and reducing surface defects by 12%
- Processed 15+ Engineering Change Requests (ECRs) with SolidWorks/AutoCAD, cutting design-to-production errors by 9%
- Supported FMEA for FRP pultrusion, identifying 10+ failure modes and initiating risk reduction actions to stabilize production quality
- Conducted daily GEMBA walks with supervisors to identify shop floor bottlenecks, driving an 8% efficiency gain

ACADEMIC PROJECTS

Design and Fabrication of a Topology-Optimized C-Clamp | Northeastern University **Jan 2025 – May 2025**

- Executed topology optimization in ANSYS under a 100 N load, reducing part volume to 32.6% while maintaining simulated peak stress within PLA allowable limits
- Fabricated via FFF additive manufacturing within a 6 g limit and ±0.2 mm tolerance, validating FEA predictions through destructive mechanical testing up to 0.63 kN

Robotic Arm Manipulation System for Automated Object Pickup | Northeastern University **Sep 2024 – Dec 2024**

- Developed a 6-DOF robotic manipulator in MATLAB for automated object pickup and material handling, implementing trajectory planning, inverse kinematics, and vision feedback
- Implemented real-time object tracking using MATLAB Computer Vision Toolbox, improving robotic manipulation accuracy through image segmentation and motion analysis

Design and Manufacturing of Conveyor System for Quality Analysis of Gears | SPPU Pune **Jun 2022 – May 2023**

- Developed an AI-powered conveyor inspection system, achieving 90%+ defect detection and reducing false rejects by 15%
- Integrated MES/SCADA systems with Power BI dashboards, enabling real-time defect analytics and improving throughput by 30% in alignment with Industry 4.0 practices

SKILLS

- **Manufacturing & Process:** Process Validation, DFM/DFA, Injection Molding, BOM Management, FRP Lamination
- **Continuous Improvement:** CAPA, NCMR, SPC, FAI, LPA, Gage R&R, Scrap Reduction, Lean Manufacturing, Six Sigma, 5s
- **Problem Solving:** Root Cause Analysis, 8D, Fishbone, 5 Whys, DMAIC, FMEA, DOE, Kaizen
- **Mechanical Design:** SolidWorks, AutoCAD, Creo, GD&T, Material Selection, Additive Manufacturing, 3D Printing
- **Analysis & Simulation:** ANSYS, FEA, CFD, Thermal Analysis, HVAC, MATLAB, Robot Mechanics, Aerodynamics
- **Data & Engineering Software:** Minitab, SQL, Python, Power BI, Tableau, MES/SCADA, Advanced Excel, SAP, Data Visualization
- **Certifications:** Certified Lean Green Belt (IISE), Certified SolidWorks Associate (CSWA), Certified SolidWorks Sustainability Associate

EDUCATION

Master of Science in Mechanical Engineering | Northeastern University | Boston, USA **Sep 2024 – May 2026**

Bachelor of Engineering in Mechanical Engineering | Savitribai Phule Pune University | Pune, India **Jul 2019 – May 2023**